

Quantification: 543_3

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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image

measure	value
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lacuna count	10
interior lacuna count	9
frame-edge lacuna count	1
gap bridges added (detection)	14

Interior lacunae only: mean, median and sample SD (n = 9)

measure	kind	mean	median	sd	units
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area_px2	lacuna shape	2187.22	2174	598.8	px^2
major_axis_length_px	lacuna shape	82.3704	87.6934	10.0353	px
minor_axis_length_px	lacuna shape	36.7931	35.2412	11.1805	px
aspect_ratio	lacuna shape	2.387	2.2947	0.6949	unitless
eccentricity	lacuna shape	0.8811	0.9	0.0775	unitless
solidity	lacuna shape	0.8856	0.9295	0.0959	unitless
roots_count	headline	6.8889	7	1.9003	count
ring_length_r30_px	headline	313.556	300	91.1017	px
ring_length_r60_px	reported	989.889	958	155.043	px
owned_length_px	ownership-dependent	1435.15	1516.24	827.008	px
edge_count	ownership-dependent	47.1111	50	28.7943	count
mean_edge_length_px	ownership-dependent	30.6592	28.9741	4.7005	px
perimeter_px	normalised	236.382	238.108	53.1676	px
ring_area_r30_px2	normalised	9046.22	8595	1273.77	px^2
ring_area_r60_px2	normalised	22951.3	21925	2828.56	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9957	1	0.0128	unitless
ring_density_r30	normalised	0.0342822	0.0358209	0.00656847	px^-1
ring_density_r60	normalised	0.0431716	0.044117	0.00439813	px^-1
roots_per_100px_perimeter	normalised	2.9693	2.4763	0.8884	per 100 px
ring_attached_length_r30_px	network v2	220.889	241	107.032	px
ring_attached_length_r60_px	network v2	507.333	627	230.455	px
ring_length_w_r30_px	network v2	351.779	333.764	104.395	px
ring_length_w_r60_px	network v2	1119.62	1099.92	173.647	px
sholl_crossings_r10	network v2	7.7778	8	3.3082	count
sholl_crossings_r20	network v2	13.2222	13	2.9907	count
sholl_crossings_r30	network v2	16	16	2.1794	count
ring_width_mean_r30_px	width	5.0947	4.9833	0.3084	px

Per field

measure	value	units
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analysed_area_px2	1.02836e+06	px^2
total_skeleton_length_px	46420	px
canalicular_length_density_per_px	0.0451398	px^-1
junction_count	698	count
junction_density_per_px2	0.00067875	px^-2
lacuna_count	10	count
lacunae_per_px2	9.72e-06	px^-2

skeleton_component_count	711	count
mean_component_length_px	65.2883	px
median_component_length_px	14	px
field_length_density_w_per_px	0.0511037	px^-1
field_density_without_flagged_per_px	0.0451398	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	5.7797	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	8	px
width_pixels_used	46400	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels with real signal under them: bridged pixels, the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2174	80.8669	35.2412	2.2947	0.9	0.9518	0.4109	79.3965	270.898	False
2	3447	90.3677	63.9486	1.4131	0.7066	0.674	0.4223	197.059	916.902	False
3	1883	69.2617	34.8694	1.9863	0.864	0.9448	0.5742	250.833	340.47	False
4	2052	69.2015	39.4228	1.7554	0.8219	0.9112	0.0622	388.671	369.939	False
5	2266	88.1613	35.5691	2.4786	0.915	0.9021	0.1324	473.05	745.191	False
6	1344	87.6934	23.0299	3.8078	0.9649	0.7764	0.2206	545.19	876.866	False
7	1683	71.0123	31.3735	2.2634	0.8971	0.9355	-0.0518	634.68	198.315	False
8	2203	91.2092	31.4053	2.9043	0.9389	0.9295	0.0832	734.301	635.44	False
9	2633	93.5592	36.2778	2.579	0.9218	0.9454	0.0457	780.576	481.495	False
10	531	28.7881	24.8252	1.1596	0.5063	0.9332	0.5942	1012.28	184.834	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	5	249	958	724.352	25	28.9741	4.7619
2	9	506	1333	1420.81	50	28.4162	4.9736
3	7	300	926	2529.89	80	31.6236	5.5208
4	10	348	934	2573.68	94	27.3796	4.9833
5	7	336	995	1754.89	57	30.7875	5.4732
6	6	277	1053	1516.24	53	28.6082	4.786
7	4	172	758	96.3553	4	24.0888	4.7529
8	8	348	1040	1601.53	43	37.2448	5.2694
9	6	286	912	698.583	18	38.8102	5.3308
10	1	106	451	43.2426	1	43.2426	4.3271

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	210.066	8595	21715	1	0.9616	0.0289703	0.044117	2.3802
2	363.446	12097	29071	1	1	0.0418285	0.0458533	2.4763

3	184.794	7941	19515	1	1	0.0377786	0.0474507	3.788
4	205.965	8333	20507	1	1	0.0417617	0.0455454	4.8552
5	241.907	9380	23807	1	1	0.0358209	0.0417944	2.8937
6	248.25	8529	21925	1	1	0.0324774	0.0480274	2.4169
7	191.965	7943	21525	1	1	0.0216543	0.0352149	2.0837
8	238.108	9212	24077	1	1	0.0377768	0.0431947	3.3598
9	242.936	9386	24420	1	1	0.0304709	0.0373464	2.4698
10	95.3909	3354	9525	0.6102	0.5725	0.0316041	0.0473491	1.0483

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	121	275	282.321	1099.92	5	13	18
2	410	735	574.6	1504.37	13	19	18
3	241	739	333.764	1057.93	8	11	16
4	267	627	384.463	1065.18	11	15	16
5	287	640	374.149	1127.32	8	13	14
6	128	343	309.764	1176.56	4	14	19
7	73	99	189.924	850.281	3	8	12
8	295	687	395.706	1165.38	10	14	16
9	166	421	321.321	1029.62	8	12	15
10	26	45	112.184	501.304	1	3	9

